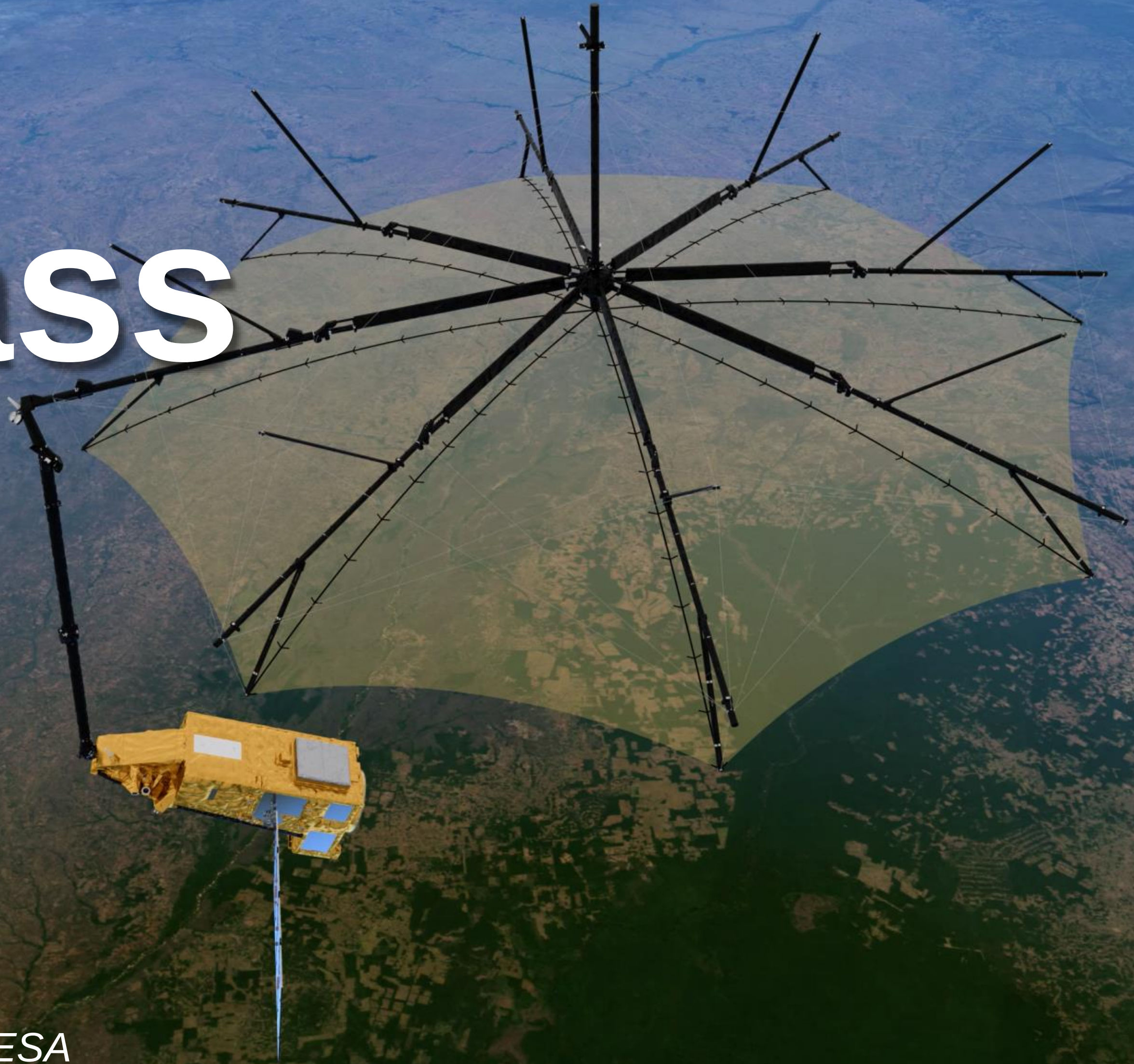
Biomass

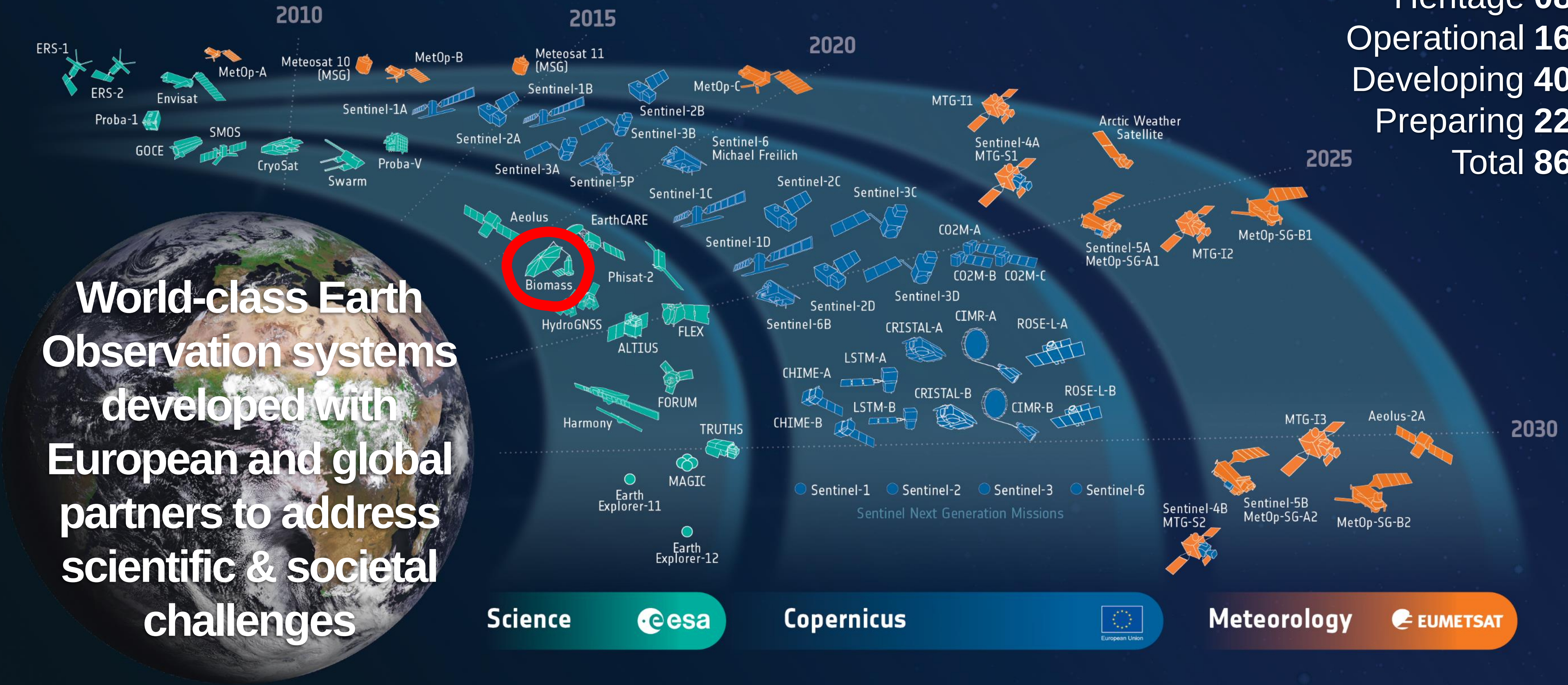


Klaus Scipal
Biomass Mission Manager , ESA

ESA's Earth Observation Missions

Satellites

Heritage **08**
Operational **16**
Developing **40**
Preparing **22**
Total **86**



World-class Earth Observation systems developed with European and global partners to address scientific & societal challenges

Forest impact us in many ways

Forests are important for **biodiversity**, as a natural **safeguard**, to regulate the **climate and weather**, as an **economic resource**, for **cultural** and **health** benefits.

Fate of anthropogenic CO_2 emissions (2014 – 2023)

Sources = Sinks

90%
9.7 ± 0.5 GtC/yr

10%
1.1 ± 0.7 GtC/yr

48%
5.2 ± 0.02 GtC/yr

30%
3.2 ± 0.9 GtC/yr

27%
2.9 ± 0.4 GtC/yr

- Land Use Change (**source**) and land uptake (**sink**) have the largest uncertainties in the global carbon budget.
- Land Use Change has a relative uncertainty of 64%!
- This reflects uncertainty in both loss and gain of **biomass**.

GLOBAL CARBON project

Difference between estimated sources & sinks = -0.5 GtC/yr indicating bias in one or more of the estimates.

Source: Friedlingstein et al 2024; Global Carbon Project 2024

How do we know how much carbon is in a tree

What information do we need?

1. We need estimates of **forest biomass (AGB), height and disturbances**.

2. The crucial **information need is in the tropics**

   deforestation (~95% of the Land Use Change flux)

   regrowth (~50% of the global biomass sink)

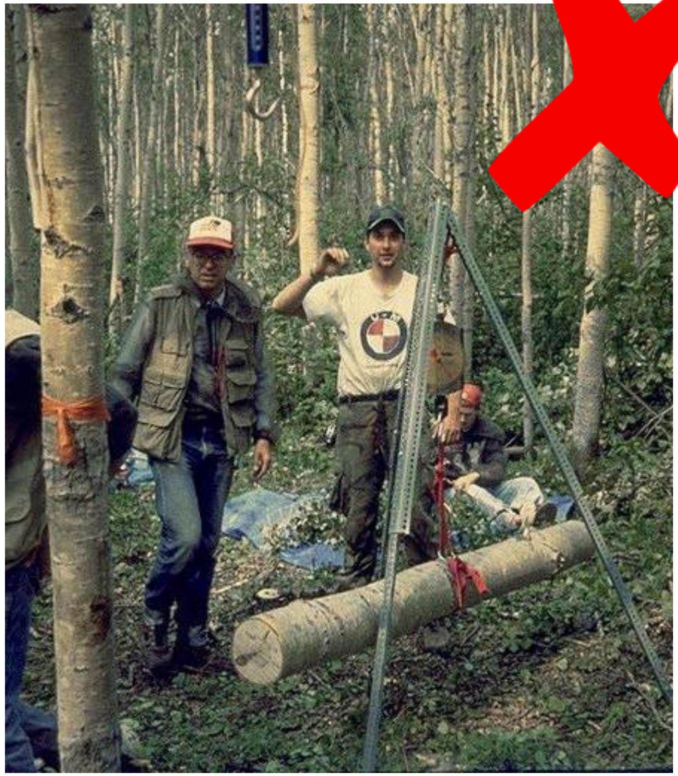
   Where we have least information and knowledge.

3. Biomass measurements are needed where the changes occur and at the **effective scale of change**: hectare scale.

4. Measurements are needed **wall-to-wall** with **repeated measurements** over multiple years to identify deforestation and regrowth.

5. A biomass accuracy of 20% at the hectare scale, **comparable to ground-based observations**.

How to measure the weight of a tree?



Tree allometry links biomass to

$$AGB = \rho \cdot \left(D/2\right)^2 \cdot H$$

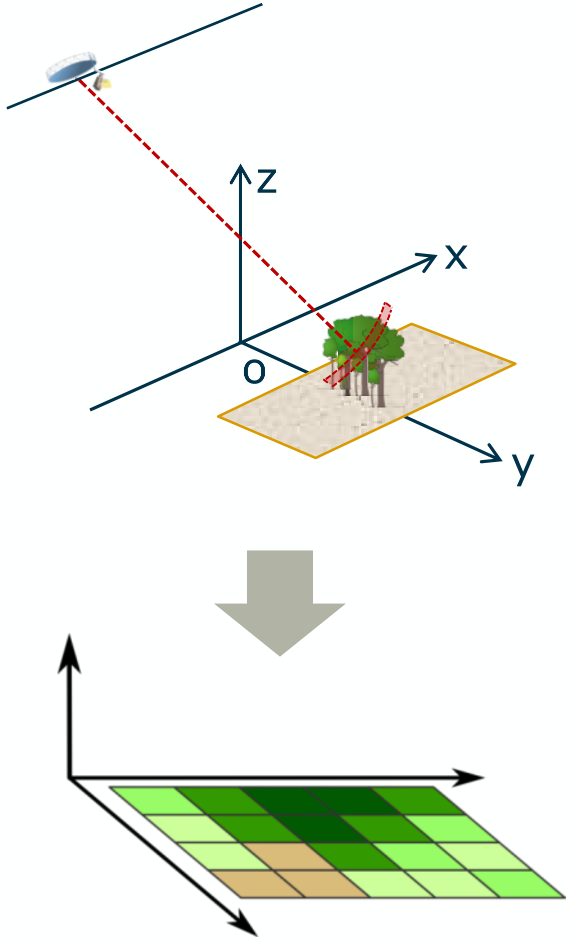
Wood density

Diameter

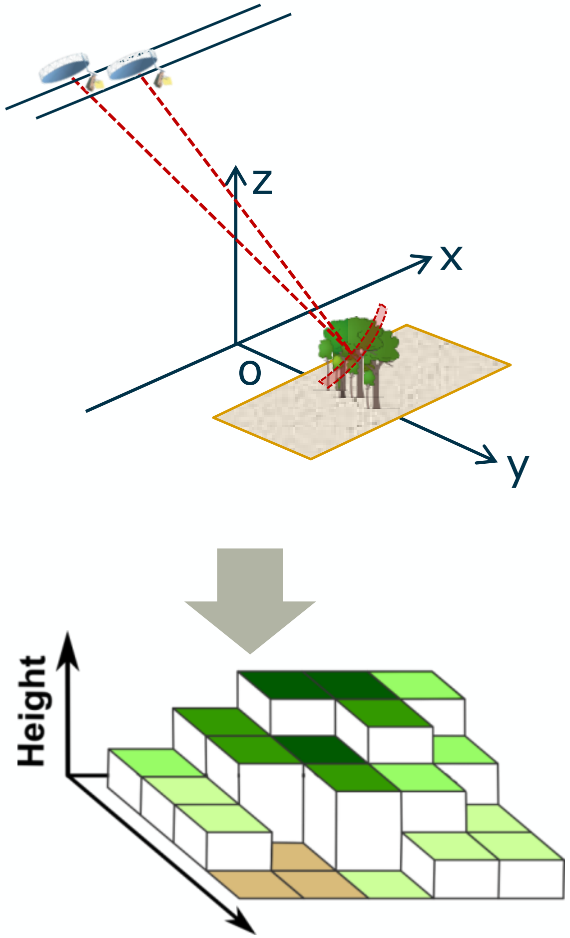
Height

Synthetic Aperture Radar contains structure information

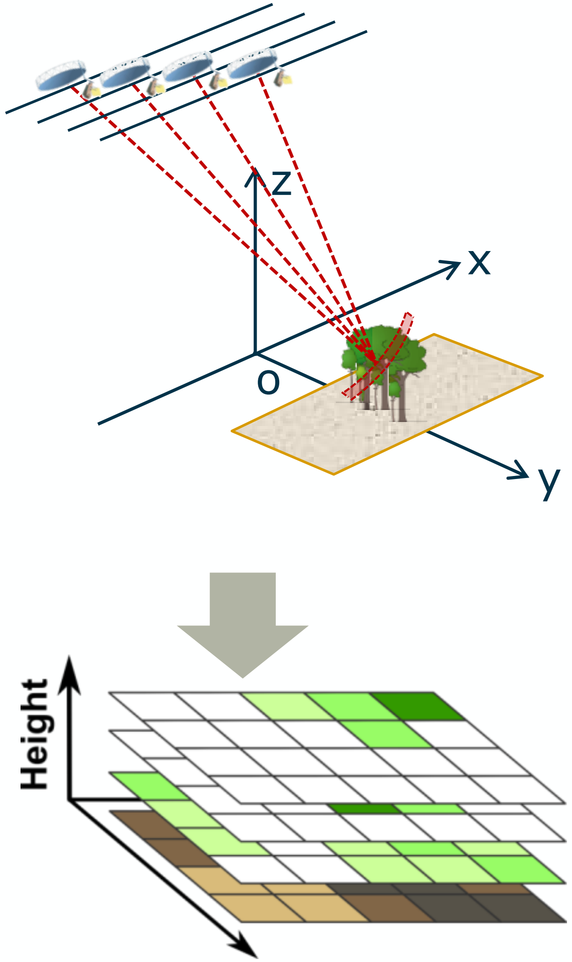
PolSAR
(SAR Polarimetry)



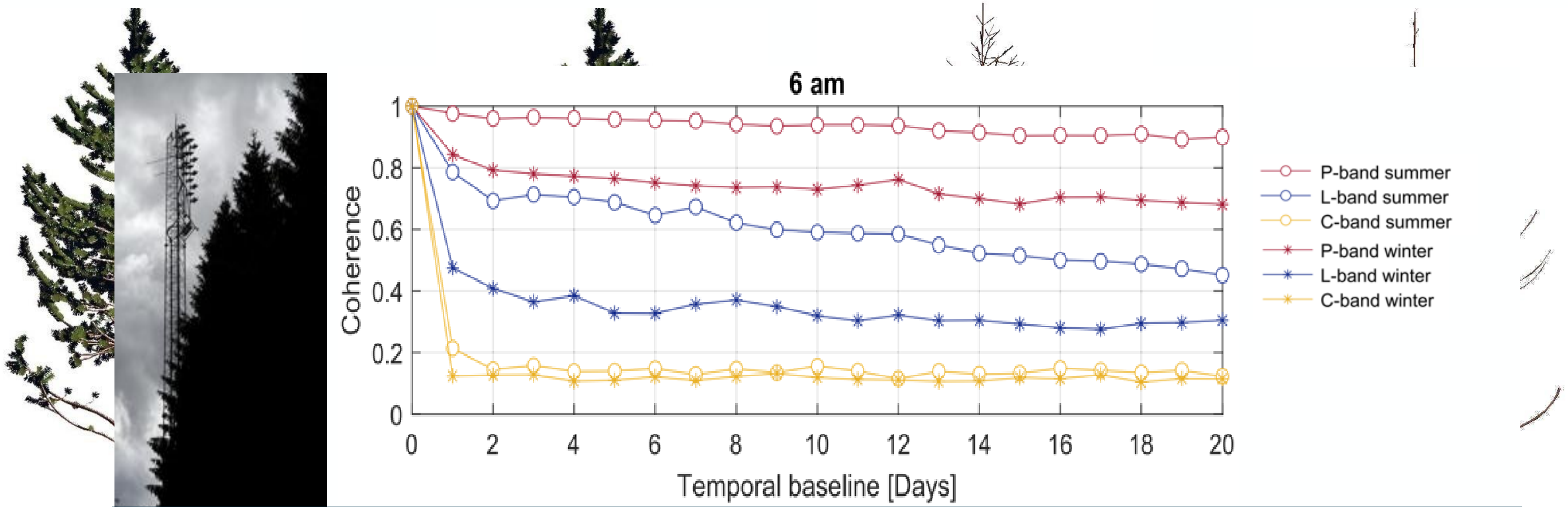
PolInSAR
(Polarimetric SAR Interferometry)



TomoSAR
(SAR Tomography)



Choice of frequency



ESA BorealScat experiment. Median temporal coherence over temporal baselines of multiples of one day.
From Monteith and Ulander, TGRS, 2021

Pinus Nigra

X-band
λ= 3 cm

L-band
λ= 27 cm

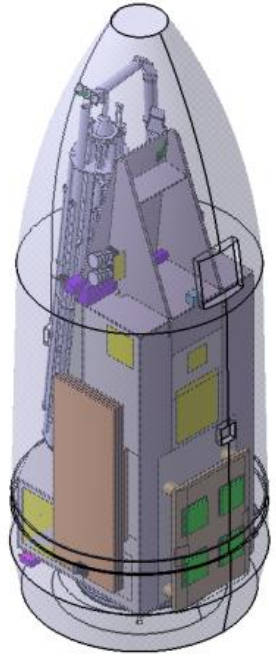
P-band
λ= 70 cm

P-band 'sees' the trunk and (big) branches, provide 'more direct' information on woody above ground biomass. In addition it is coherent with time enabling repeat pass interferometry.

Biomass Mission Requirements

System Specifications

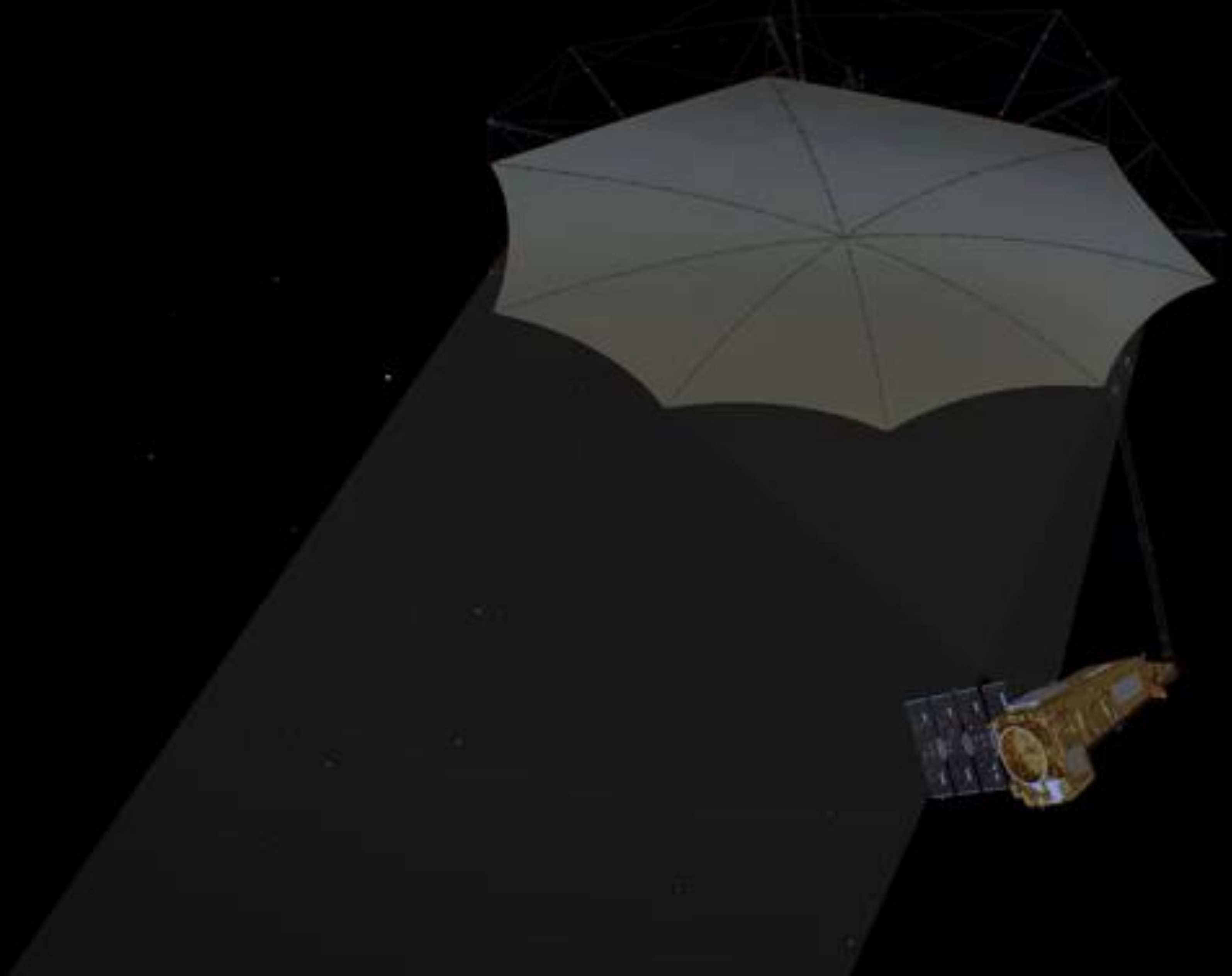
Instrument	Polarimetric SAR operating at 465 MHz (~70 cm wavelength)
Channel Imbalance	≤ -25 dB, Tx and Rx combined
Cross-Talk	≤ -30 dB
Radiometric bias	≤ 0.3 dB
Radiometric stability	≤ 0.5 dB
Noise Equivalent Sigma Nought	≤ -27 dB
Total Ambiguity Ratio	≤ -18 dB
Spatial resolution (SLC), range and azimuth	~ 59 m x 8 m
Residual phase error, standard deviation	≤ 10 deg, over pulse travel and data take time (12 min)
Peak to Sidelobe Ratio	≤ -16 dB
Integrated Sidelobe Ratio	≤ -9 dB
Geo-location accuracy	Better than 25 m
Dynamic range	-30 dB to 5 dB
Swath Width	~ 50 km



Orbit

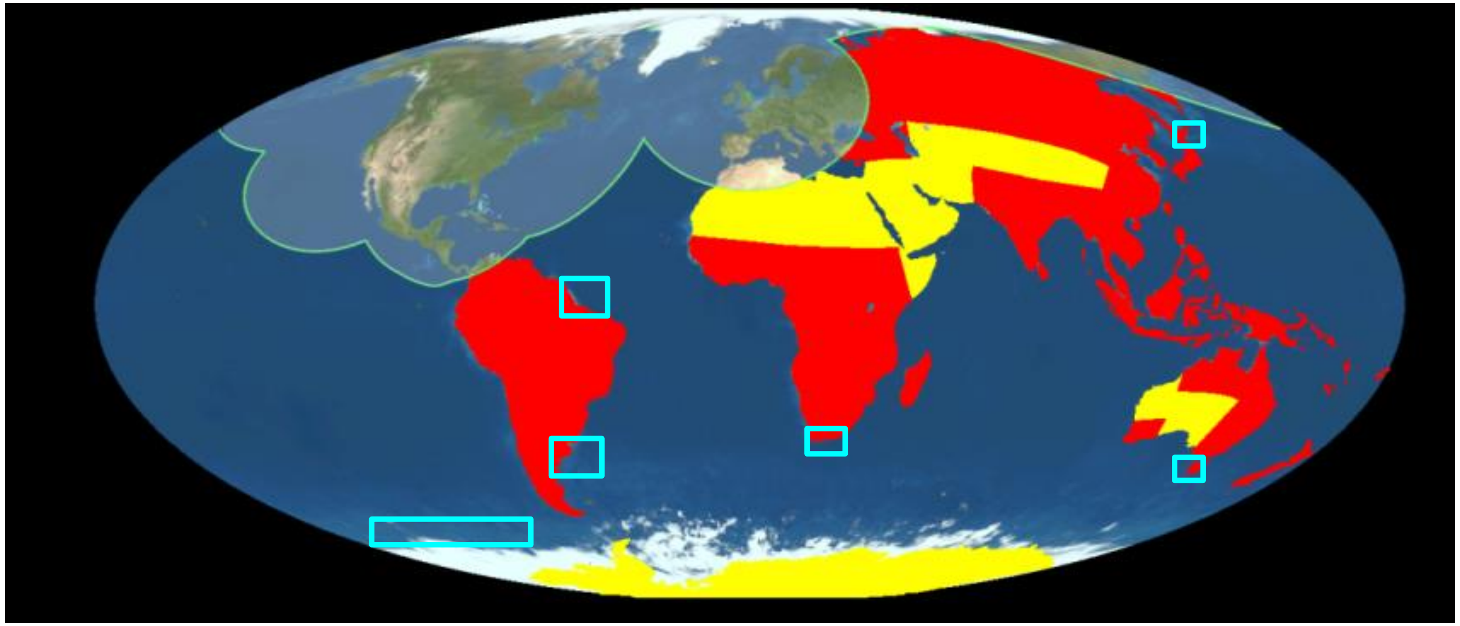
Orbit Type	Sun synchronous, dawn dusk
Baseline	East-west drift of 4.2 km

How Biomass achieves global coverage



Coverage

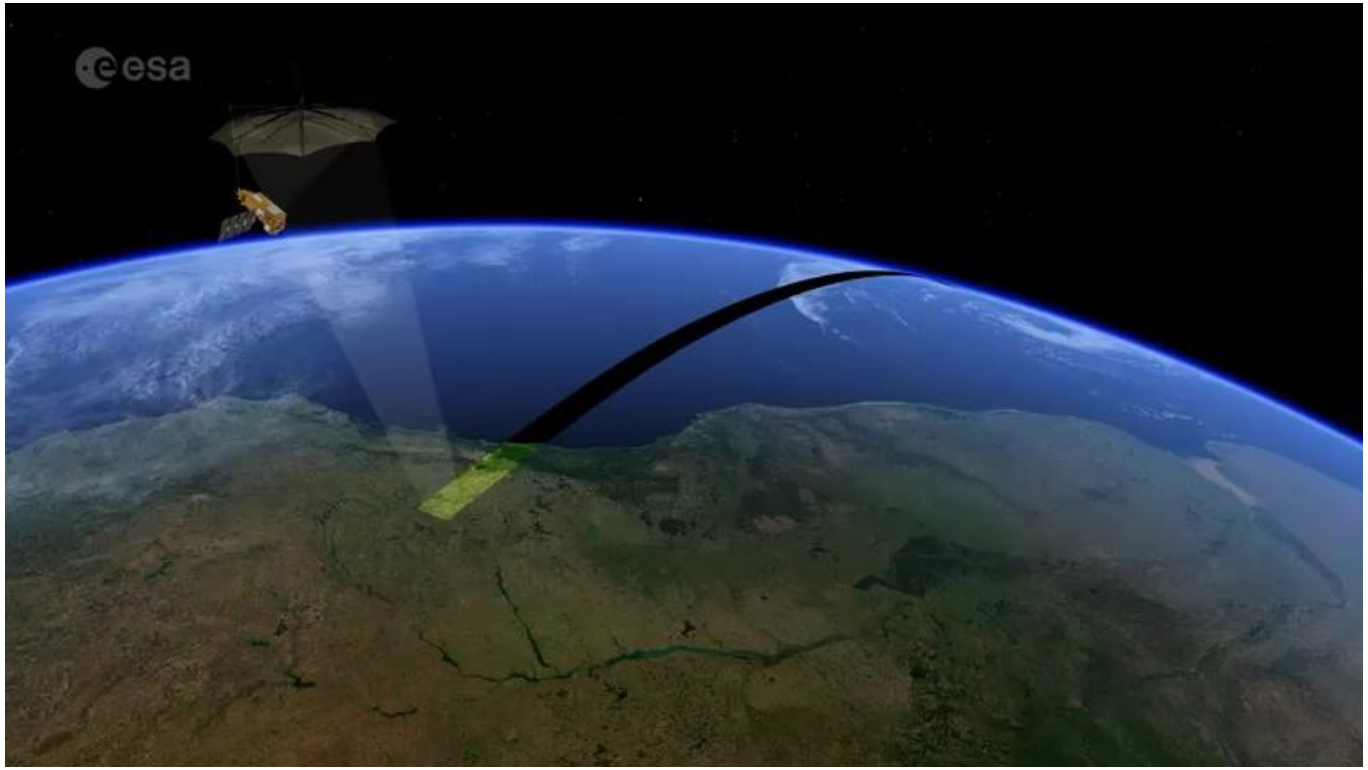
- Systematic Acquisitions for forested land (red area)
- Global coverage in ~17 months (TOM phase) and ~8 months (INT phase).
- Best effort acquisitions for non forested areas (yellow + ocean/sea ice ROIs)
- Acquisition mask restricted by US Space Objects Tracking Radar (SOTR)



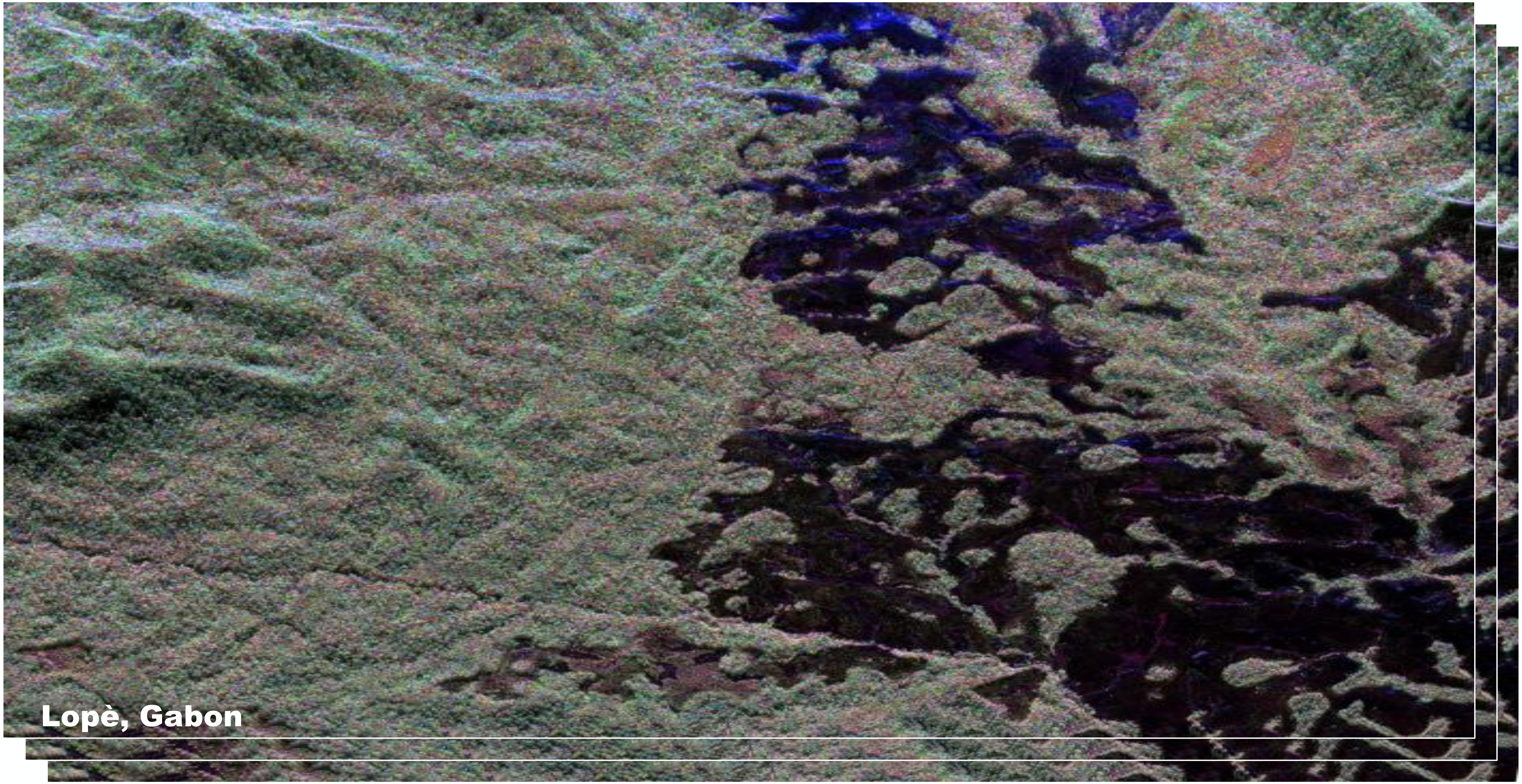
(Red = Primary objective coverage mask, Yellow = Secondary objective coverage mask)

The fascinating journey from a radar pulse to tons of wood per hectare

- A complex 100 W radar pulse, 30 microseconds long, will be sent out 3000 times per second
- Reflected pulses as low as 0.0000000000001 Watts can be used
- Then comes the complex Synthetic Aperture Radar processing
- Here the scientists enter: their algorithms can convert the “radar Images” to maps of “tons of wood” per hectar



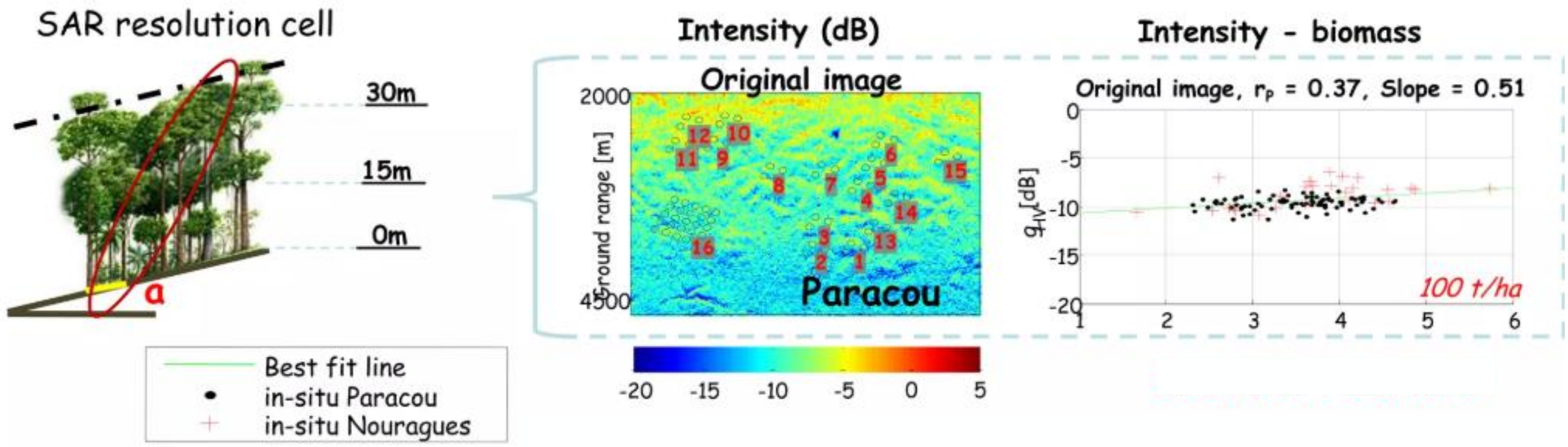
Challenge – Retrieve forest biomass and height



Tropical Forest as seen by DLR‘s P-band F-SAR

HH+VV HV HH-VV

How to retrieve biomass from intensity

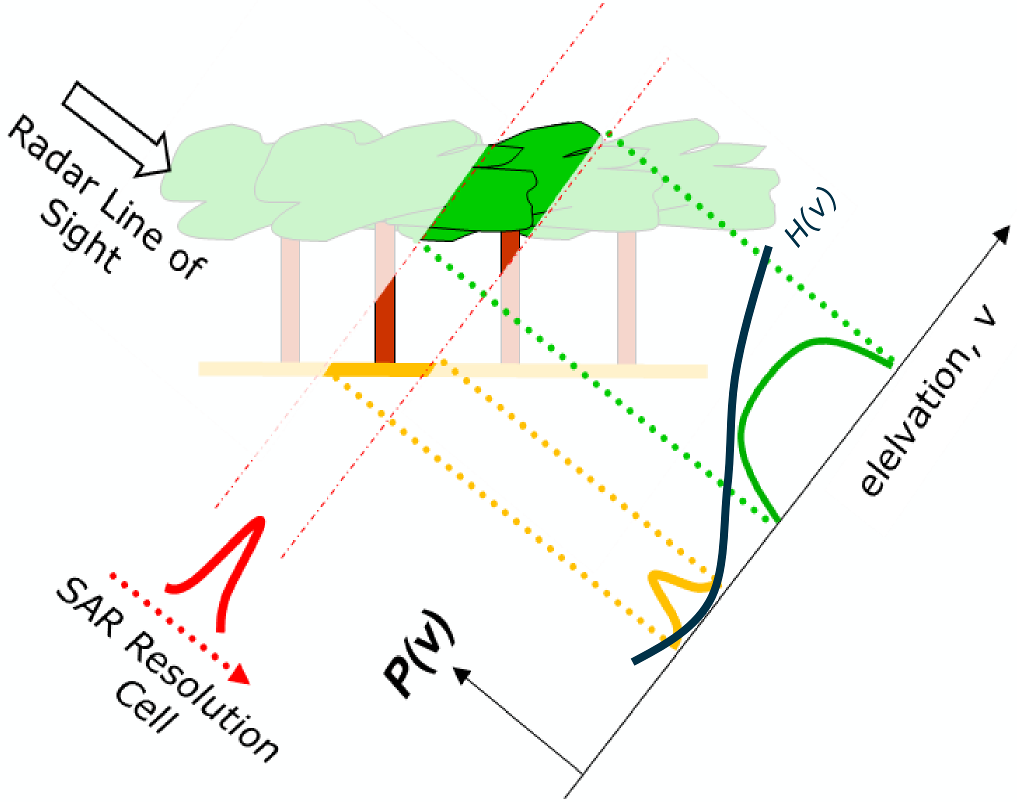


The Biomass AGB retrieval: Interferometric ground notching

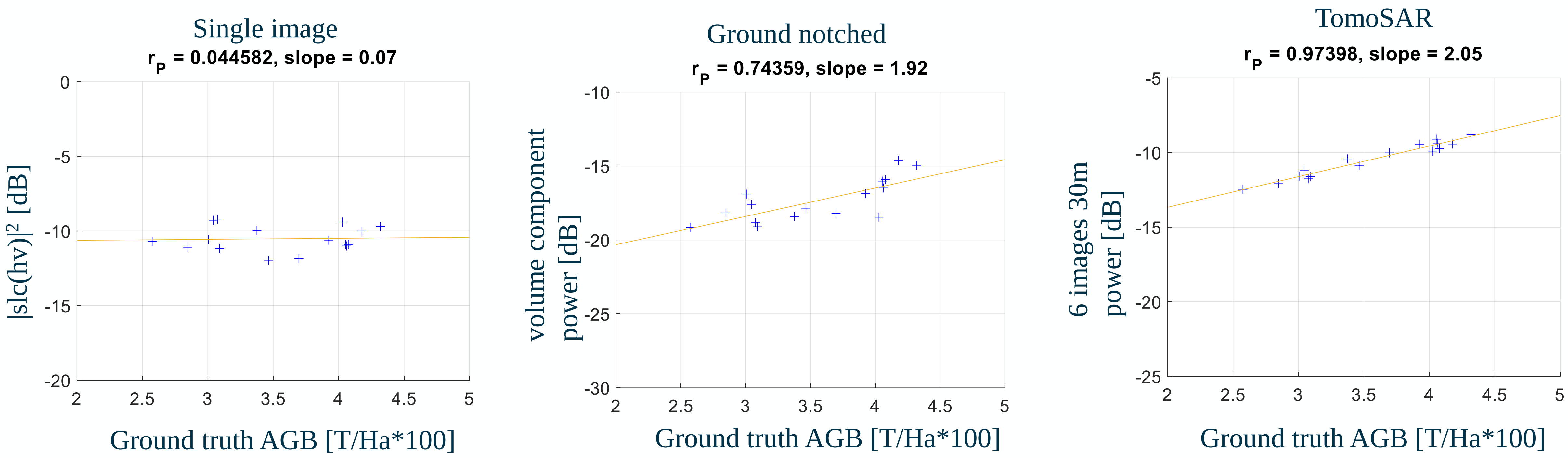
Ground cancellation: the ground cancellation technique was developed to preserve the advantages of SAR Tomography during Mission lifetime.

Idea: cancel out ground scattering by taking the difference between two phase calibrated SLC BIOMASS images.

Principle: SLC = projection of modulated target reflectivity along elevation.

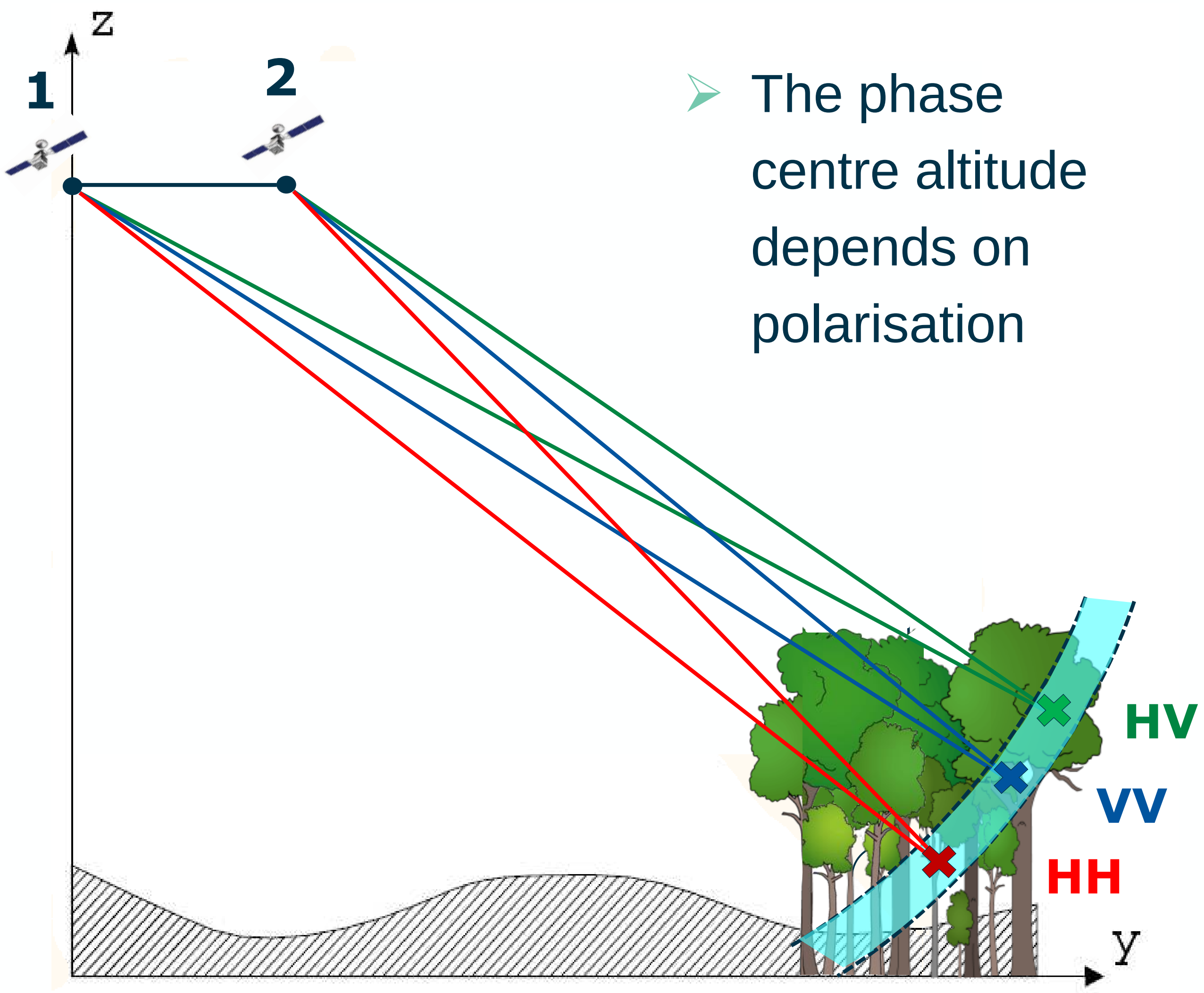


AGB vs TropiSAR backscatter

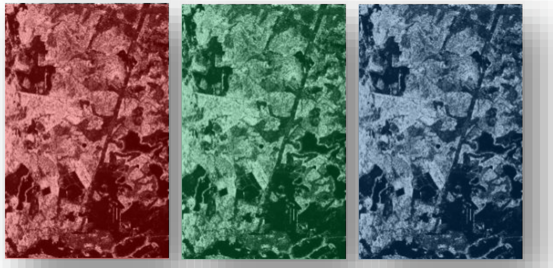


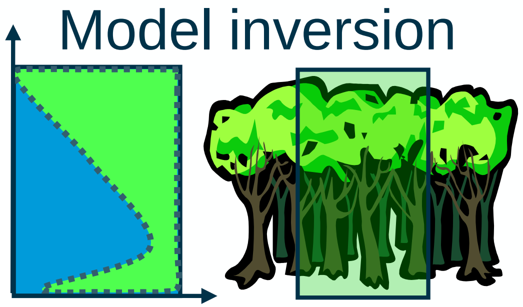
Ground rejection greatly improves the sensitivity and correlation

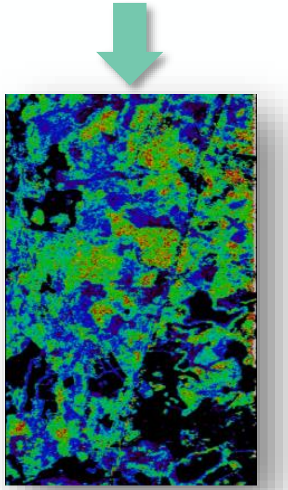
Polarimetric Interferometric SAR



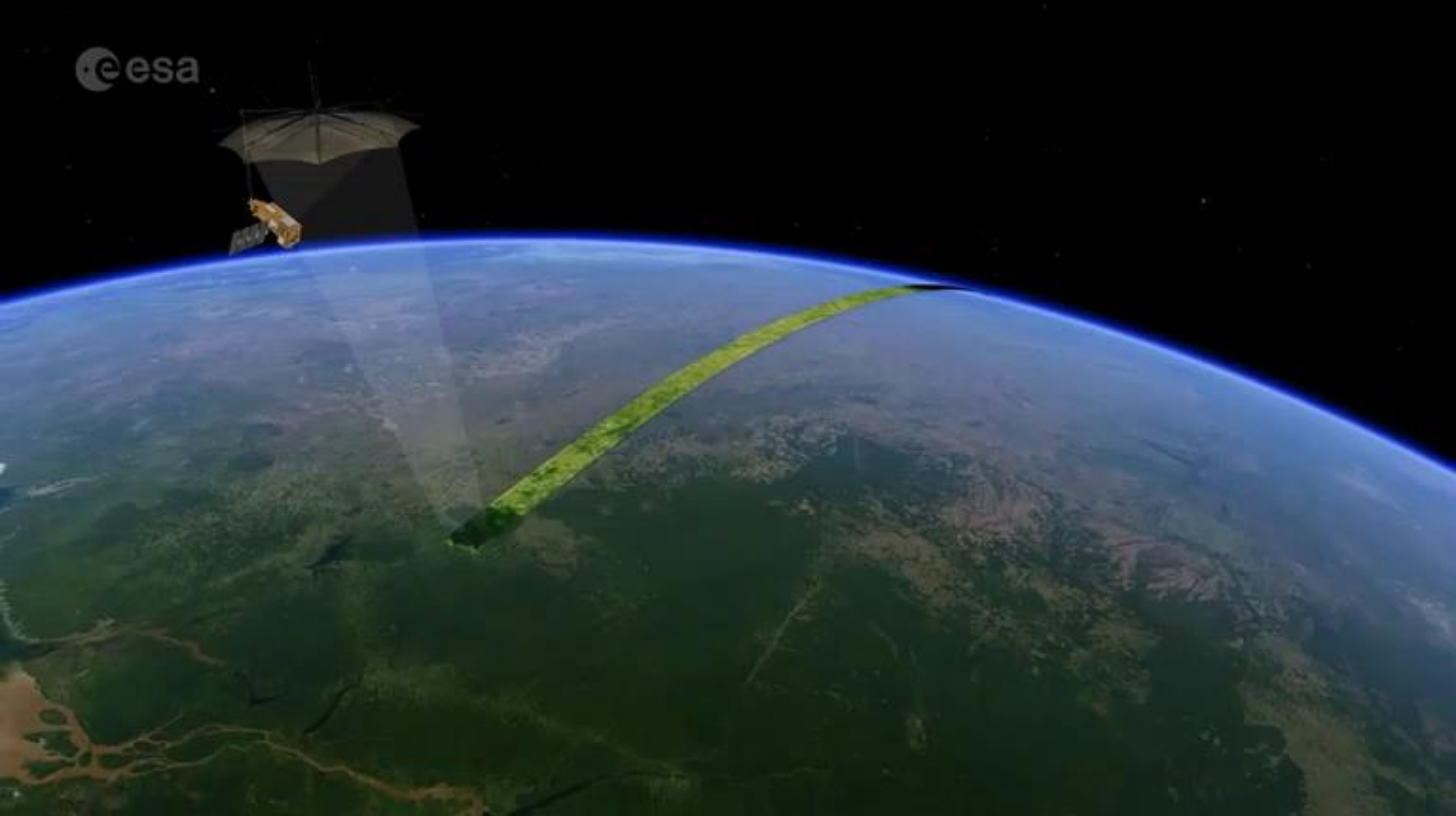
- The phase centre altitude depends on polarisation







Vegetation height



Biomass Products

Above-ground biomass (tons/hectare)

- 200 m resolution
- accuracy of 20%, or 10 t ha–1 for biomass < 50 t ha–1

Upper canopy height (meter)

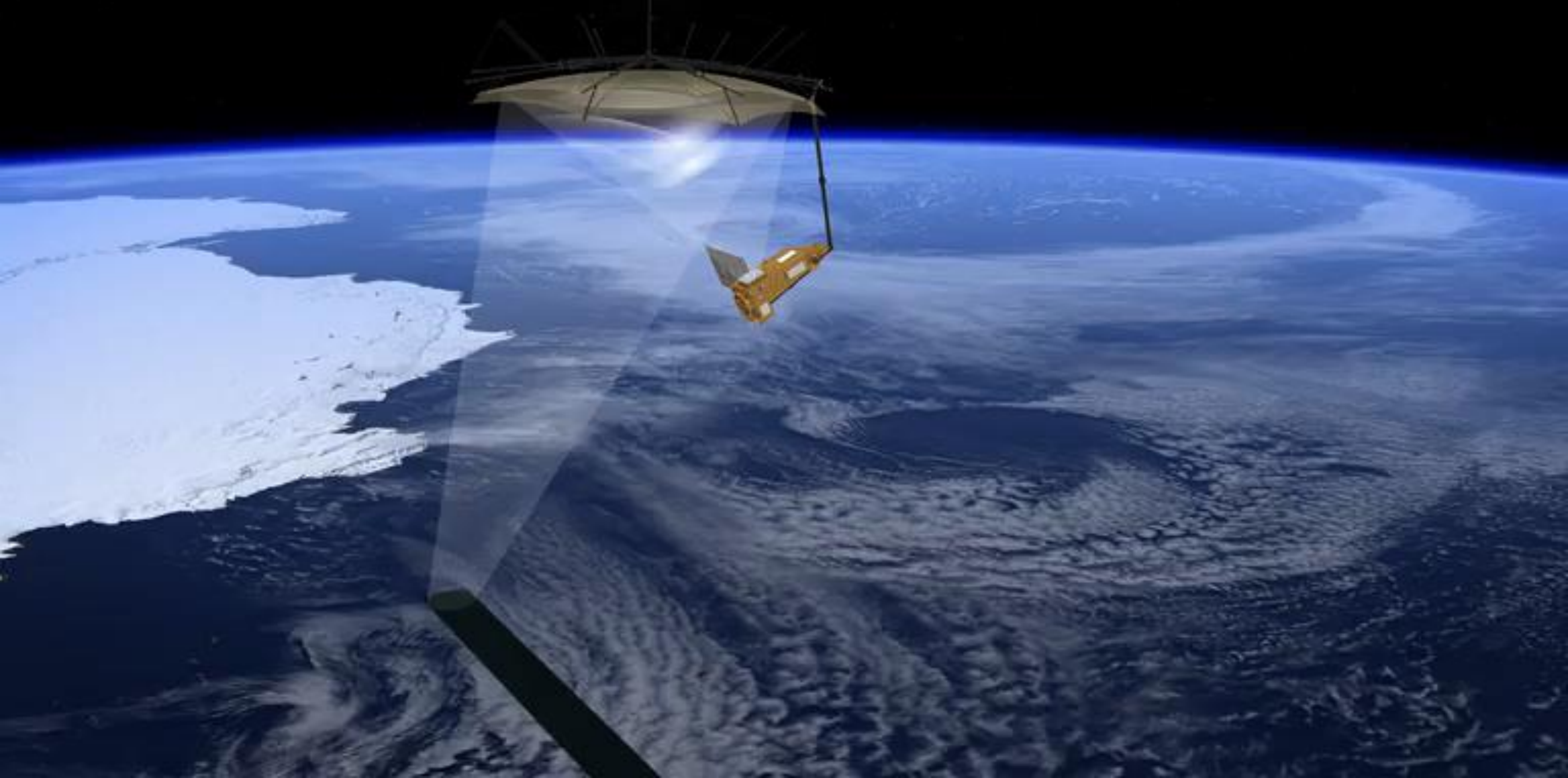
- 200 m resolution
- accuracy of 30%

Areas of forest clearing (hectare)

- 50 m resolution
- 90% classification accuracy

- 1 map every 8 months of all forested areas (excl. SOTR region)

BIOMASS beyond carbon



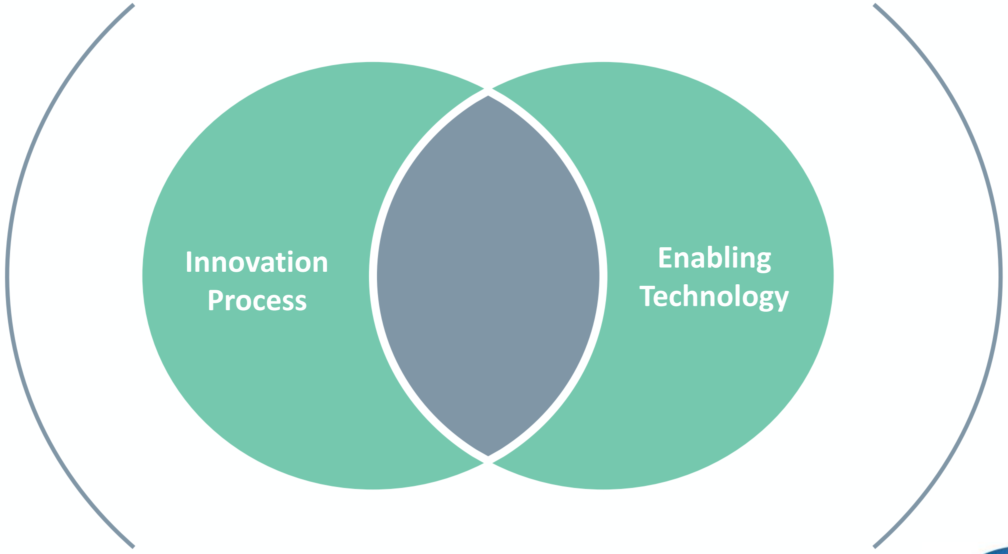
BIOMASS an Open Innovation Explorer

esa

Facilitate advanced research and services in EO and beyond, through enabling an open, digital ecosystem based on the principles of openness, inclusiveness, and innovation.

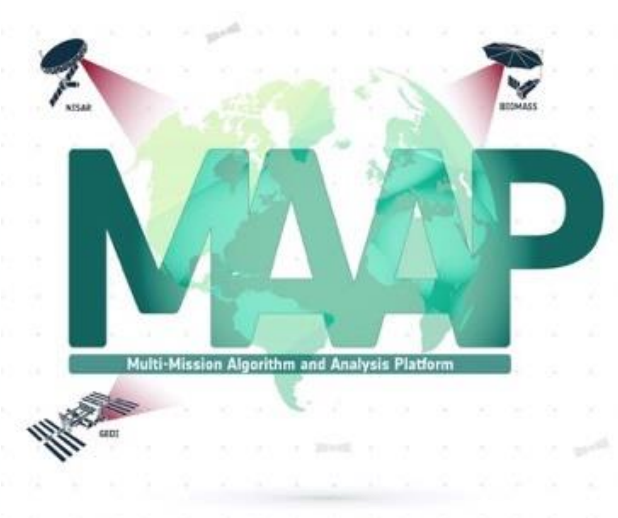
BioPAL

biopal@esa.int
biopal.org
github.com/BioPAL



esa
European Space Agency

NASA



BIOMASS

GEDI ECOSYSTEM LIDAR

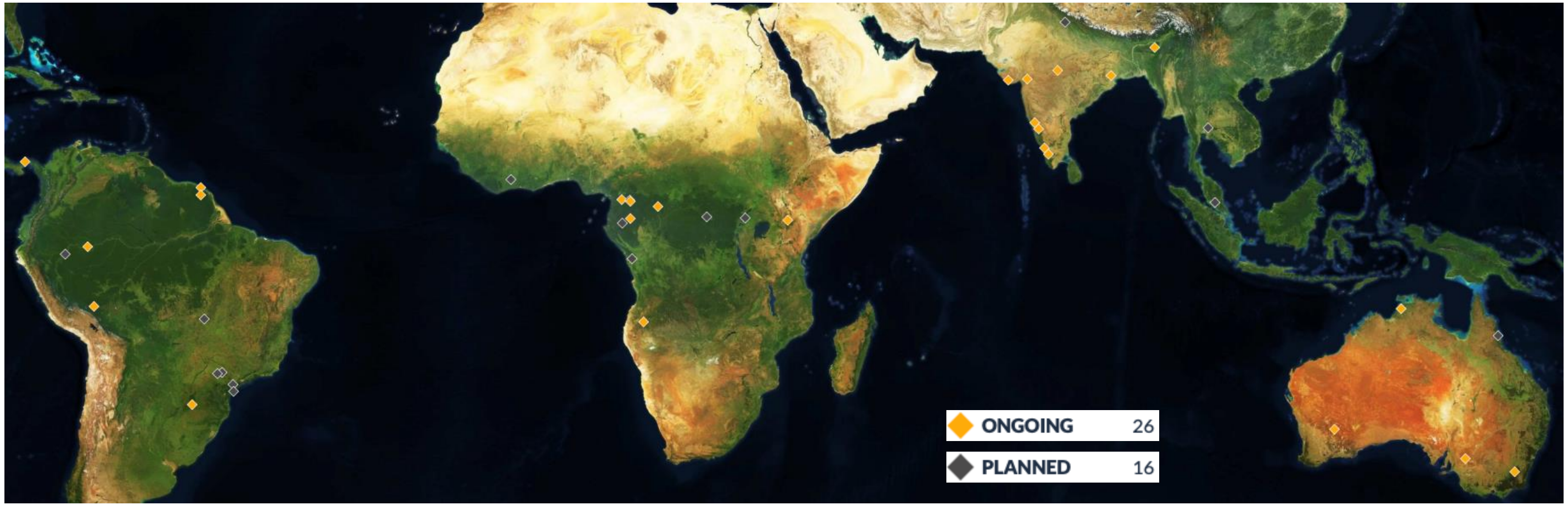
NISAR NASA - ISRO SAR Mission

Enabling an equitable pantropical validation network

Collecting ground truth in the tropics remains a challenge!

- ➢ **requires skills, expertise and resources**

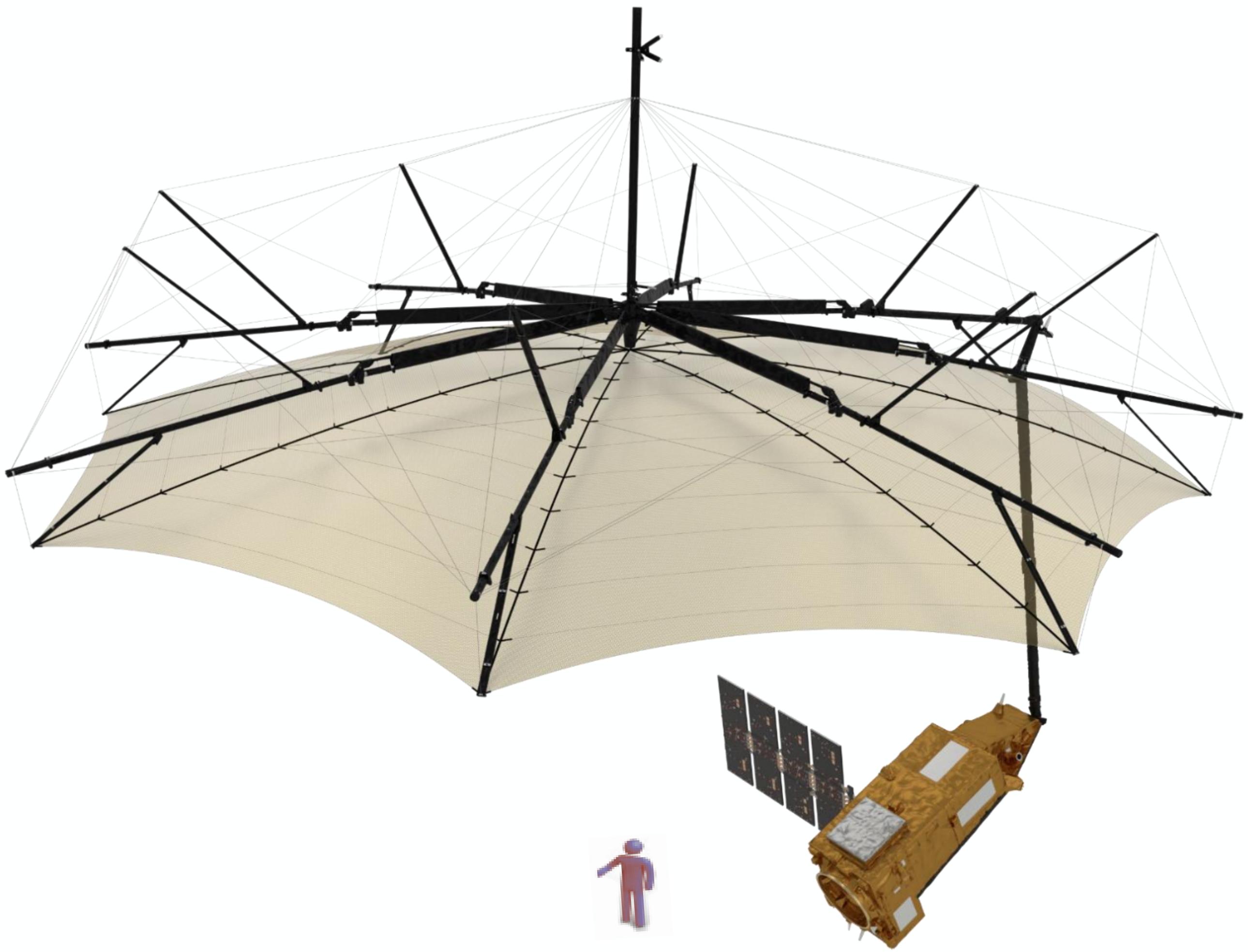
The recent GEO-TREES initiative finances the collection of high-quality forest information for EO forest product validation building on existing forest research sites



https://geo-trees.org

What it takes to build a satellite like Biomass

- **20 years** from the first proposal, 9 years build and test the satellite.
- **287 M€** for the satellite (**500 M€** total cost including pre-development, launch, and operations).
- Consortium of **80+ companies** from **23 countries** from Europe + Canada, the USA and Australia.
- 1200 people involved at one point, **700+ people** involved at peak development time.



Summary – BIOMASS a true Earth Explorer

1. BIOMASS was proposed in 2005. Implementation started in Nov. 2015. The satellite will be ship on the 22nd February to Kourou where it will undergo final testing. Launch is on the 29th April 2025.

2. BIOMASS is the first P-band SAR the and first mission to systematically acquire radar tomographic data. It is a true Earth Explorer; we will face a lot of unknowns but also a lot of exciting science opportunities.

3. It is the first Earth Explorer not only sharing its data open and free but also following Open Innovation best practices.

4. The new unique vision of Earth from Biomass will extend beyond forests and into measurements of ice, sub-surface geomorphology in deserts, topography, the ionosphere, ocean ...

Thank you

ESA's Biomass Mission